FairLight: City-Wide Emergency Vehicle Route Prioritization via Envy-Free Hierarchical Multi-Agent Actor–Critic

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Abstract

Urban traffic signal control is a complex sequential decision-making problem in which decisions made at one intersection can influence traffic conditions at neighboring and downstream intersections. The problem becomes increasingly challenging as the size of the road network grows because a controller must simultaneously achieve network-level coordination, local responsiveness, and computational scalability. Reinforcement learning has emerged as a promising approach for adaptive traffic signal control, while multi-agent reinforcement learning (MARL) allows control to be distributed among individual intersections. However, fully decentralized approaches may lack sufficient global context, whereas centralized approaches become difficult to scale.

This paper proposes FairLight, a Hierarchical Region-Aware Multi-Agent Reinforcement Learning framework for Fair and Adaptive Traffic Signal Control. The framework introduces a regional abstraction between network-level coordination and intersection-level decision-making. The traffic network is partitioned using graph community detection. Regional traffic states are subsequently processed by a high-level Meta-Policy using spatial and temporal representations, while low-level intersection agents use local observations, neighboring information, and regional guidance to select signal actions.

A central feature of the proposed framework is the explicit treatment of fairness as a traffic-control objective. Conventional evaluation based only on average delay, average travel time, or throughput can hide excessive delays experienced by a smaller group of vehicles. The proposed framework therefore incorporates vehicle-level waiting-time distribution, maximum waiting time, tail waiting time, and Jain's fairness index into optimization and evaluation. A multi-objective reward function jointly considers traffic efficiency, fairness, and emergency responsiveness.

The framework also treats region construction as part of the research problem rather than merely a preprocessing step. DBSCAN, Louvain, Leiden, and hybrid region construction methods are investigated to determine how the choice of regional abstraction affects downstream traffic-control performance.

Keywords: Traffic Signal Control, Multi-Agent Reinforcement Learning, Emergency Vehicle Routing, Community Detection, Louvain Algorithm, Fairness-Aware Optimization, Hierarchical Reinforcement Learning

1. Introduction

Urban traffic signal control is a complex sequential decision-making problem where decisions at one intersection propagate to neighboring and downstream intersections. Traffic demand varies dynamically with time, incidents, weather, and travel patterns, making static or rule-based controllers insufficient for real-world conditions. Adaptive systems such as SCOOT introduced traffic-responsive coordination [2], while reinforcement learning approaches have since enabled data-driven signal control [11]. Multi-agent reinforcement learning (MARL) further distributes control across intersections, improving scalability, though fully decentralized approaches risk optimizing locally at the expense of network-wide coordination [20, 19].

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Beyond aggregate congestion, traffic signal decisions simultaneously affect travel time, delay, queue formation, throughput, fuel consumption, and emissions. Critically, evaluating controllers solely through aggregate metrics such as average waiting time conceals how service is distributed among individual vehicles. Consider two controllers with vehicle waiting times $W_A = [10, 10, 10, 10, 10, 100]$ and $W_B = [25, 25, 25, 25, 25, 25]$: Controller A achieves a lower mean delay yet inflicts severe starvation on one vehicle. This motivates fairness as a first-class control objective rather than a post-hoc evaluation criterion.

A further dimension of real-world uncertainty that existing MARL frameworks largely overlook is the presence of emergency vehicles (EMVs). Most traffic signal control approaches, including hierarchical methods such as HiLight [24] and cooperative approaches such as CoLight [17], assume homogeneous vehicle populations and optimize purely for aggregate efficiency. In practice, ambulances, fire engines, and police vehicles impose stochastic, high-priority demands on the network that cannot be resolved by ordinary signal timing. Delayed EMV passage has measurable consequences: studies report approximately 700 annual fatalities attributable to late ambulance responses, and an estimated annual mean of 4,500 ambulance-involved crashes in the United States [35]. EMVLight demonstrated that jointly optimizing EMV routing and adaptive signal control within a multi-agent actor-critic framework substantially reduces emergency response times [31], but does not address hierarchical city-wide coordination or fairness. These considerations establish EMV responsiveness as a necessary component of any realistic traffic management framework, not an optional extension. At the same time, providing unrestricted priority to an EMV may excessively delay ordinary traffic, potentially transferring congestion to other parts of the network rather than resolving it. EMV prioritization should therefore not be treated as an isolated intersection-level problem: it requires coordination across the surrounding traffic network while maintaining acceptable service for non-emergency vehicles.

Hierarchical reinforcement learning addresses the scalability–coordination tradeoff by separating long-horizon strategic decisions from short-horizon local control [22, 23]. HiLight is particularly relevant here, employing a Meta-Policy with Transformer-based spatial and LSTM-based temporal modelling to coordinate decentralized Sub-Policy agents across large networks [24]. However, its region partitioning is treated as a preprocessing artifact rather than a control-relevant variable, and it neither incorporates EMV priority nor vehicle-level fairness.

This paper proposes FairLight: City-Wide Emergency Vehicle Route Prioritization, an Envy-Free Hierarchical Multi-Agent Actor-Critic framework that jointly addresses these limitations. The framework partitions the traffic network using graph community detection methods (DBSCAN, Louvain, Leiden, and hybrid variants), treating region construction as a research variable whose effect on downstream control performance is explicitly investigated. A hierarchical coordination layer issues regional guidance to local intersection agents, while a multi-objective reward function simultaneously optimizes traffic efficiency, vehicle-level fairness, and EMV responsiveness. To the best of our knowledge, no existing framework jointly integrates hierarchical MARL, topology-aware regional partitioning, envy-free fairness optimization, and dynamic multi-EMV routing within a unified city-wide architecture.

In summary, this paper makes four contributions. First, we treat regional partitioning as a first-class, empirically investigated design variable – evaluating DBSCAN, Louvain, Leiden, and hybrid community-detection/density-clustering combinations – rather than a fixed manual preprocessing step, and show that automatically detected regions adapt to network topology and demand density in ways a fixed manual grid partition cannot (Section 6.3). Second, we extend single-EMV preemption to a city-wide, multi-EMV routing mechanism (Section 3.2) that injects a time-decaying, route-anticipatory bias into the hierarchical state representation, and report a controlled on/off comparison (Table 1) isolating its contribution over purely local EMV preemption. Third, we formulate traffic efficiency, vehicle-level fairness, and EMV responsiveness as an explicit, user-configurable multi-objective reward (Eq. 23–24) rather than fixed, hand-tuned constants, and expose accident/incident robustness (Section 3.3) as an evaluated training condition rather than an untested assumption. Fourth, we situate this work against the scale of the problem it targets: emergency vehicle delay is associated with on the order of 700 annual fatalities and 4,500 ambulance-involved crashes in the United States alone [35], and the urban populations served by the signalized road networks this framework targets number in the millions – making EMV-aware, fairness-aware traffic control a high-population-impact problem rather than a narrow research exercise.

2. Related Work

Conventional traffic signal control has primarily relied on analytical optimization and rule-based methods for determining optimal signal timings. Webster's signal timing formulation introduced one of the earliest mathematical

models for fixed-cycle traffic signal optimization, while later adaptive traffic control systems such as SCOOT and SCATS dynamically adjusted signal timings using detector-based traffic measurements and predefined control strategies [1, 2, 3, 4]. Max-pressure control further improved network throughput by selecting signal phases according to queue pressure differences between incoming and outgoing links [5]. Although these approaches are computationally efficient and have been successfully deployed in practice, they rely heavily on predefined traffic models and handcrafted optimization rules. Consequently, they struggle to adapt to highly dynamic urban environments involving unpredictable congestion, heterogeneous intersections, accidents, and emergency vehicle requests.

To improve adaptability, several optimization and machine learning techniques have been introduced for traffic signal control. Evolutionary optimization methods such as Genetic Algorithms optimize signal timing plans through iterative search, while Fuzzy Logic Controllers employ expert-defined rules to respond to uncertain traffic conditions [6, 7]. Later studies adopted machine learning techniques including Support Vector Machines, Artificial Neural Networks, and Long Short-Term Memory (LSTM) networks to predict traffic flow and congestion using historical observations [8, 9, 10]. Although these methods improve prediction accuracy and optimization quality, they primarily estimate future traffic states rather than learning sequential traffic control policies. Furthermore, they require extensive historical training data and cannot continuously adapt to rapidly changing traffic conditions.

Recent advances in deep reinforcement learning have significantly improved adaptive traffic signal control by enabling traffic controllers to learn directly through interaction with the traffic environment. IntelliLight demonstrated the effectiveness of Deep Q-Networks for adaptive traffic signal control at isolated intersections [11]. PressLight integrated max-pressure traffic theory into deep reinforcement learning to maximize traffic throughput while reducing handcrafted reward engineering [12]. FRAP modeled traffic control as a phase competition problem and exploited traffic phase symmetry to improve learning efficiency [13]. AttendLight later introduced an attention-based architecture capable of generalizing across heterogeneous intersections with varying lane configurations [14]. MetaLight focused on transferring learned policies across unseen intersections [15], while Advanced-XLight demonstrated that effective traffic state representation can significantly influence performance in reinforcement-learning-based traffic signal control [16]. Despite these advances, existing deep reinforcement learning methods primarily optimize local traffic efficiency and do not jointly consider emergency vehicle prioritization, fairness, or large-scale hierarchical coordination.

To improve cooperation among multiple intersections, graph neural networks and multi-agent reinforcement learning have received increasing attention. CoLight introduced graph attention networks to enable communication among neighboring intersections for cooperative traffic signal control [17]. MPLight further improved scalability by introducing parameter sharing across reinforcement learning agents, allowing efficient deployment over city-scale transportation networks [18]. HiLight extended this concept by proposing a hierarchical reinforcement learning framework consisting of Meta-Policy and Sub-Policy agents for coordinating thousands of intersections efficiently [24]. CityFlow provides a large-scale multi-agent reinforcement learning environment designed specifically for city traffic scenarios and supports evaluation of scalable traffic-control algorithms [21]. Although these methods substantially improve scalability and inter-agent cooperation, their primary objective remains maximizing traffic efficiency. They neither address emergency vehicle routing nor incorporate fairness-aware resource allocation across different regions of a city.

HiLight's hierarchical Meta-Policy/Sub-Policy formulation [24], like most of the methods discussed above, operates under a closed-world assumption: it is trained and evaluated on recurrent congestion patterns and does not model non-recurrent disruptions such as emergency vehicle preemption or traffic incidents. This omission is not merely conceptual. Poorly coordinated EMV preemption has been shown to increase average delay for cross-traffic by 18–32 seconds per vehicle per preemption event, with these localized delays cascading into network-wide congestion [32]. Even well-established preemption strategies carry a measurable efficiency cost: EMVLight's own evaluation shows that traditional preemption techniques increase average non-EMV travel time by approximately 10% relative to unpreempted baselines [31]. These results indicate that emergency vehicle disruption is a first-order cost of operating a real traffic network, not a negligible edge case that a closed-world controller can safely ignore. Motivated by this, our framework treats EMVs and accident events as first-class constraints on the control problem rather than external events layered on top of it, at the cost of an increase in average travel time and delay relative to a disruption-free benchmark—a cost we treat as the price of a tighter approximation to real-world operating conditions.

Emergency vehicle routing and traffic signal pre-emption have also received considerable research attention. Conventional emergency routing methods generally employ shortest-path algorithms such as Dijkstra's algorithm, A*,

and various time-dependent shortest path formulations to determine optimal routes under static or partially dynamic traffic conditions [25, 26, 27, 28, 29]. More recent studies have introduced reinforcement learning-based routing strategies capable of adapting to changing traffic environments [30]. EMVLight jointly optimizes emergency vehicle routing and adaptive traffic signal control using multi-agent actor–critic reinforcement learning, significantly reducing emergency response times while maintaining acceptable traffic efficiency for ordinary vehicles [31]. Subsequently, hierarchical multi-agent actor–critic methods have been proposed to further improve emergency routing through coordinated traffic signal control [33]. However, existing emergency traffic management approaches generally assume decentralized coordination and do not explicitly consider fairness, topology-aware regional decomposition, or simultaneous optimization for multiple emergency vehicles operating within large urban transportation networks. Visual sensing-based approaches have further demonstrated that incorporating real-time distance measurement and vehicle counting into traffic management can substantially improve EMV scheduling and intersection clearance [35].

Fairness has recently emerged as an important objective in intelligent transportation systems. FairMARL introduced a fairness-aware multi-agent reinforcement learning framework based on generalized social welfare functions to reduce disparities among intersections while maintaining competitive traffic efficiency [34]. Although fairness-aware optimization improves equitable resource allocation, existing approaches primarily focus on balancing intersection delays and do not integrate emergency vehicle routing, hierarchical coordination, or topology-aware traffic management into a unified framework.

Overall, significant progress has been achieved in analytical optimization, deep reinforcement learning, graph-based coordination, emergency vehicle routing, and fairness-aware traffic signal control. However, these objectives have largely been investigated independently. Classical optimization methods lack adaptability, deep reinforcement learning approaches prioritize traffic efficiency over fairness, graph-based coordination methods do not explicitly support emergency response, emergency vehicle routing methods overlook hierarchical city-wide coordination, and fairness-aware methods ignore emergency traffic management.

To the best of our knowledge, no existing framework jointly integrates hierarchical multi-agent reinforcement learning, topology-aware regional partitioning, envy-free fairness optimization, and dynamic multi-emergency vehicle routing within a unified city-wide traffic management architecture. The proposed framework, FairLight, addresses these limitations by introducing an Envy-Free Hierarchical Multi-Agent Actor–Critic architecture that simultaneously optimizes traffic efficiency, fairness, emergency response, and scalable coordination across heterogeneous urban transportation networks.

3. Proposed Framework

The proposed framework, FairLight, is organized around three complementary methods that together address scalability, emergency responsiveness, and equity in city-wide traffic signal control. First, the road network is automatically partitioned into coherent subregions using graph-based community detection (the Louvain algorithm), removing the need for manual, expert-driven region design and allowing the hierarchical Meta-Policy/Sub-Policy architecture to scale to arbitrarily large networks. Second, the framework extends single-vehicle emergency routing to a multi-EMV setting, in which several emergency vehicles may simultaneously request priority across overlapping parts of the network, requiring explicit conflict resolution and coordinated corridor clearing. Third, the framework treats fairness and accident-induced congestion as first-class control objectives rather than side effects, so that the learned policy neither systematically starves individual approaches nor collapses when unexpected incidents disrupt normal traffic flow. The remainder of this section describes each of these three methods in turn.

3.1. Clustering Based on Louvain

In the proposed framework, manually dividing the road network into precise subsections is impractical for real-world scenarios due to the large number of roads and intersections involved. Therefore, the road network is modeled as a weighted graph $G = (V, E)$, where intersections are represented as nodes and road segments as edges.

To automatically identify suitable subsections, the Louvain community detection algorithm is applied to the graph. The algorithm partitions the network into communities by maximizing modularity, which measures the difference between the observed connectivity within communities and the connectivity expected in a corresponding random network:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \tag{1}$$

where A_{ij} is the edge weight between nodes i and j, k_i and k_j are their weighted degrees, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether the nodes belong to the same community.

A higher modularity indicates stronger community structure, allowing the identified communities to serve as automatically generated road-network subsections. Each subsection is assigned a sub-policy, while a higher-level meta-policy coordinates the behavior of these sub-policies across the complete network. This provides a scalable alternative to manual decomposition and enables hierarchical policy optimization over complex road networks. Figure 1 illustrates this on Cologne8: intersections before any regional abstraction, and the same intersections colored by detected cluster (Section 6.3 discusses why a manual grid partition cannot adapt to a network's actual connectivity the way this does).

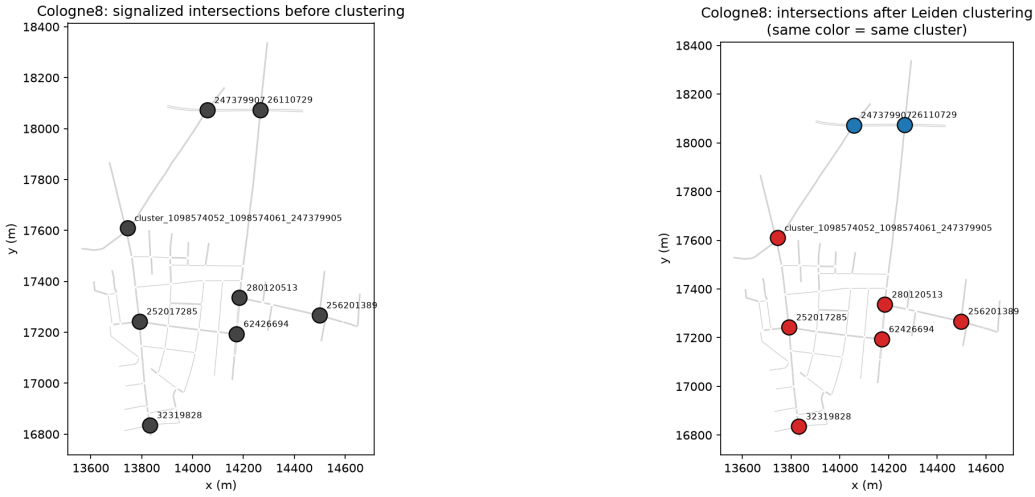


Figure 1: Cologne8 signalized intersections before regional clustering (left) and after Leiden clustering, same color = same cluster (right).

3.2. Multiple EMV Vehicle Routing Across the City

Prior emergency-vehicle (EMV) prioritization approaches in the traffic-signal-control literature, including the single-agent preemption strategies discussed in Section 2, are almost exclusively formulated around a single emergency vehicle traversing the network at any given time. This assumption breaks down in dense urban environments, where it is common for multiple ambulances, fire trucks, and police vehicles to be simultaneously active, often converging on overlapping segments of the road network from different origins and with different destinations. Extending EMV prioritization to a genuinely multi-vehicle setting therefore requires the framework to reason jointly about several time-critical routes rather than optimizing each one in isolation.

We propose to model the multi-EMV routing problem as a joint optimization over a set of active emergency vehicles $\mathcal{E} = \{e_1, e_2, \ldots, e_K\}$, each associated with a current position, a destination, and a priority class (for example, life-threatening medical emergency versus structural fire response). At each decision step, the framework must determine (i) a preferred route for every active emergency vehicle through the signalized network, and (ii) how to allocate green-phase priority at any intersection that lies on more than one emergency vehicle's preferred route. This second requirement is the central new difficulty relative to single-EMV routing: when two emergency vehicles approach the same intersection from conflicting directions within a short time window, granting unconditional priority to both is physically impossible, and the controller must instead arbitrate based on factors such as estimated time-to-intersection, remaining distance to destination, and priority class.

We envision extending the existing Meta-Policy/Sub-Policy hierarchy so that each active emergency vehicle contributes an additional, time-decaying bias term to the regional and intersection-level state representations along its

anticipated route, rather than introducing a wholly separate routing module. Conceptually, this bias term would increase in magnitude as an emergency vehicle approaches a given intersection and decay once it has passed, so that the learned Sub-Policy naturally favors clearing a path ahead of an approaching emergency vehicle while reverting to its ordinary efficiency/fairness trade-off once the vehicle has moved on. Coordination across subregions, needed when an emergency vehicle's route crosses several Louvain-derived subregions in sequence, would be handled by the Meta-Policy, which is already responsible for issuing cross-regional guidance.

Resolving conflicts between multiple simultaneously active emergency vehicles is expected to require an explicit arbitration rule layered on top of this bias mechanism — for instance, prioritizing the vehicle with the least estimated time-to-intersection, breaking ties by priority class, and bounding the maximum delay that can be imposed on a lower-priority emergency vehicle so that no single vehicle is perpetually deprioritized in favor of others. The corresponding evaluation methodology would extend the single-EMV response-time metrics used elsewhere in this work (e.g., average and 95th-percentile emergency waiting time) to the multi-vehicle case by additionally reporting the spread of response-time improvements across all active emergency vehicles, so that the framework can be shown to accelerate every emergency vehicle's passage and not merely the one considered by the controller to be of highest priority.

Finally, we note that a multi-EMV setting also changes how the framework's fairness objective (Section 3.3) should be interpreted: granting priority to emergency vehicles necessarily imposes additional waiting time on ordinary traffic, and this additional cost must be distributed across the network rather than concentrated on whichever ordinary vehicles happen to be sharing a road segment with an active emergency-vehicle corridor at a given moment. This interaction between multi-EMV routing and the fairness reward is left as an open design question to be resolved in the full version of this section.

3.3. Fairness and Accidental Congestion Control

Traffic-signal controllers optimized purely for network-wide efficiency metrics, such as average travel time or average delay, can produce policies that are efficient in aggregate yet highly inequitable at the level of individual approaches or vehicles. A controller may, for instance, learn to sacrifice one chronically low-priority approach in order to keep throughput high on the network's dominant corridors, resulting in a small subset of vehicles experiencing persistent, disproportionate waiting times even while the network-wide average delay looks acceptable. The fairness component of the proposed framework is intended to make this trade-off explicit and controllable rather than an incidental by-product of reward design.

We plan to quantify fairness along several complementary axes: the variance of waiting times across approaches at a given intersection, the maximum (tail) waiting time experienced by any single approach, and Jain's Fairness Index computed over per-approach or per-vehicle service levels. Unlike a single scalar efficiency term, these measures are each sensitive to a different failure mode — variance captures persistent imbalance between approaches, the tail-waiting-time penalty specifically targets starvation of the worst-served approach, and Jain's index provides a normalized, scale-independent summary of overall equity that can be tracked and compared across scenarios and partitioning methods. An "envy" term, comparing each intersection's service level against that of its neighboring intersections, is further intended to discourage locally greedy policies that improve one intersection's performance strictly at a neighbor's expense — directly relevant to the Louvain-based regional decomposition of Section 3.1, since poorly balanced subregions could otherwise induce exactly this kind of neighbor-level inequity.

A second, largely orthogonal concern addressed by this subsection is robustness to accidental congestion: collisions, breakdowns, and other unplanned incidents that suddenly and sharply reduce the effective capacity of one or more road segments. Conventional traffic signal controllers are typically trained and evaluated exclusively under nominal demand conditions, and so may respond poorly when a lane or approach is unexpectedly blocked, continuing to allocate green time as though the obstructed segment retained its usual capacity. We intend to address this by exposing the controller to a representative distribution of injected incidents during training — varying in location, duration, and severity — so that the learned Sub-Policy and Meta-Policy learn to detect the resulting shift in local traffic statistics (e.g., a sudden rise in queue length or a collapse in mean speed on an otherwise normal approach) and reallocate green time accordingly, rather than requiring a separate, hand-engineered incident-detection and override layer.

We additionally plan to evaluate whether accident-aware training degrades the framework's fairness guarantees: an incident response that simply redirects all available green time toward the obstructed approach could, in principle, starve unrelated approaches elsewhere in the network. The evaluation methodology for this subsection will therefore

report fairness metrics (variance of waiting time, tail waiting time, Jain's index) separately for incident and incident-free simulation intervals, to confirm that the controller's response to accidental congestion does not come at the cost of the equity objectives established earlier in this section. Together with the multi-EMV routing extension of Section 3.2, this completes the three methods — scalable graph-based regional decomposition, coordinated multi-emergency-vehicle routing, and fairness-aware, incident-robust congestion control — that constitute the proposed FairLight framework.

3.4. Hierarchical Architecture Overview

The proposed FairLight framework addresses the scalability and coordination challenges of city-scale adaptive traffic signal control through a hierarchical multi-agent reinforcement learning architecture. A fully centralized controller becomes computationally expensive as the number of intersections increases, whereas independent intersection-level agents lack sufficient network-wide context to coordinate traffic across connected corridors. To address this trade-off, FairLight introduces an intermediate regional abstraction between network-level coordination and local signal control.

The framework consists of four principal components: (i) automatic regional construction, (ii) a network-level Meta-Policy, (iii) intersection-level Sub-Policies, and (iv) a safety-constrained signal execution layer. Regional construction transforms the signalized road network into coherent traffic regions, the Meta-Policy extracts global spatial-temporal information and generates network-level control targets, and the Sub-Policies translate this global guidance into local phase decisions using intersection and neighborhood information. The resulting hierarchy is trained jointly using PPO with an adversarial goal-alignment mechanism.

3.5. Traffic Network Representation and Automatic Regionalization

Manually decomposing a large urban road network into meaningful control regions is impractical because the number and connectivity of signalized intersections can vary substantially across networks. FairLight therefore represents the signalized road network as a weighted graph

$$G = (V, E), \tag{2}$$

where each vertex $v_i \in V$ represents a signalized intersection and each edge $e_{ij} \in E$ represents the connectivity between two signalized intersections. Non-signalized junctions between traffic lights are contracted during graph construction so that the resulting graph directly represents the controllable signalized-intersection topology.

The connectivity between intersections is characterized using a traffic-dependent edge weight rather than purely static topology. For an edge e, the implemented weighting function is

$$w_e = \alpha n_e + \beta \omega_e + \gamma \kappa_e n_e, \tag{3}$$

where n_e denotes the number of vehicles on the edge, ω_e represents accumulated waiting time, and

$$\kappa_e = \mathrm{clip}\left(1 - \frac{v_e}{v_e^{\max}}, 0, 1\right) \tag{4}$$

denotes the congestion ratio. The implementation uses $\alpha = 1.0, \beta = 0.3$, and $\gamma = 2.0$. These weights are averaged over a 1000-step simulation window, allowing the regional structure to reflect persistent traffic interactions rather than a transient traffic state.

The resulting weighted graph is partitioned into communities using graph community-detection methods. In particular, Louvain and Leiden identify communities by maximizing graph modularity,

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m}\right) \delta(c_i, c_j), \tag{5}$$

where A_{ij} is the weighted adjacency matrix, k_i and k_j are the weighted degrees of vertices i and j, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether the two vertices belong to the same community. DBSCAN and hybrid graph-density approaches are additionally supported, allowing regionalization itself to be evaluated as an experimental variable rather than treated as a fixed preprocessing choice.

This regionalization produces a set of regions

$$\mathcal{R} = \{R_1, R_2, \ldots, R_M\}, \tag{6}$$

which forms the intermediate abstraction used by the hierarchical controller. Each region provides aggregated traffic information to the Meta-Policy, while individual intersections within the regions remain independently controllable by the Sub-Policies.

3.6. Hierarchical Meta-Policy

The Meta-Policy provides network-level coordination by observing the temporal evolution of traffic conditions across the generated regions. For each region R_k, the regional state is represented as

$$s_k^t = \left[N_k^t, W_k^t, x_k, y_k\right] \in \mathbb{R}^4, \tag{7}$$

where N_k^t denotes the aggregate stopped-vehicle count, W_k^t denotes aggregate waiting time, and (x_k, y_k) represents the normalized spatial centroid of the region.

Because traffic conditions are inherently temporal, the Meta-Policy maintains a sliding window of $T = 20$ regional observations. A Transformer encoder processes the regional tokens jointly to capture spatial dependencies between regions. A learnable global token is introduced to obtain a network-level representation

$$F_g \in \mathbb{R}^4. \tag{8}$$

The temporal representations of individual regions are subsequently processed using an LSTM. The resulting hidden states are concatenated across the M regions and transformed through an MLP to obtain a global representation $G \in \mathbb{R}^{16}$. The Meta-Policy then generates two network-level control targets,

$$(G_w, G_q) = \mathrm{Linear}_{16\to 2}(G), \tag{9}$$

where G_w represents the desired global waiting-time level and G_q represents the desired global queue-length level.

Thus, rather than directly selecting signal phases throughout the network, the Meta-Policy establishes a global traffic-control objective that can be pursued by the lower-level intersection agents. This separation allows long-range traffic coordination to be handled at a compressed regional level while preserving local responsiveness at individual intersections.

3.7. Intersection-Level Sub-Policy

The Sub-Policy converts global guidance into local signal-control actions. At each intersection i, the local observation is represented by

$$o_i^t = \left[v_i, q_i, o_i, f_i, s_i, w_i, v_i^{avg}, p_i, c_i, d_i\right] \in \mathbb{R}^{10}, \tag{10}$$

where the components capture local traffic volume, queue length, occupancy, flow, stops, waiting time, speed, pressure, congestion, and delay. The observations are normalized using an exponential moving average and clipped to a bounded range before being supplied to the policy network.

A local multilayer perceptron first encodes the intersection state. Since traffic conditions at one intersection are influenced by neighboring intersections, the encoded representation is augmented using a Graph Attention Concat (GAC) mechanism over the intersection-adjacency graph. For a neighboring pair (i, j), the attention coefficient is computed as

$$e_{ij} = \mathrm{LeakyReLU}\left(a^T[h_i \| h_j]\right), \qquad \alpha_{ij} = \mathrm{softmax}_j(e_{ij}), \tag{11}$$

and the resulting neighborhood representation is constructed through concatenation,

$$z_i = h_i \| \alpha_{i1} h_{j_1} \| \cdots \| \alpha_{i4} h_{j_4}. \tag{12}$$

The neighborhood representation is then combined with the global Meta-Policy embedding:

$$s_i^t = \left[z_i \| F_g \right]. \tag{13}$$

This representation simultaneously provides the Sub-Policy with local traffic information, neighboring traffic context, and network-level coordination guidance.

A shared actor-critic network processes this representation and produces a probability distribution over signal phases. Because intersections may have different numbers of valid signal phases, invalid action indices are masked before the final softmax operation. Parameter sharing across intersections allows the same learned policy structure to be deployed across heterogeneous signalized intersections while retaining local observations and graph-based neighborhood information.

3.8. Joint Hierarchical Optimization

The Meta-Policy and Sub-Policies are optimized jointly. The Sub-Policy is trained using Proximal Policy Optimization (PPO) with Generalized Advantage Estimation (GAE). The temporal-difference residual is

$$\delta_t = r_t + \gamma V(s_{t+1})(1 - d_t) - V(s_t), \tag{14}$$

and the generalized advantage estimate is

$$A_t = \delta_t + \gamma\lambda(1 - d_t)A_{t+1}, \tag{15}$$

with $\gamma = 0.99$ and $\lambda = 0.95$. The PPO policy objective uses the clipped probability ratio

$$L_{\text{policy}} = -\min\left(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon)A_t\right), \tag{16}$$

where $\epsilon = 0.2$.

The hierarchical relationship is enforced through an additional goal-alignment term. Let W_{global} and Q_{global} denote the actual network-wide waiting time and queue length. The Meta-Policy is trained using

$$L_{\text{Meta}} = \text{MSE}\Big((G_w, G_q), (W_{\text{global}}, Q_{\text{global}})\Big) + \eta_1 r_g, \tag{17}$$

while the Sub-Policy receives an alignment objective encouraging the realized network state to follow the Meta-Policy's targets:

$$L_{\text{Sub}} = L_{\text{AC}} + \eta_2 \left[\beta_q(Q_{\text{global}} - G_q) + \beta_w(W_{\text{global}} - G_w)\right]. \tag{18}$$

Here, $\eta_1 = \eta_2 = 0.1$ and $\beta_q = \beta_w = 0.5$. The alignment gradient is detached from the Meta-Policy in the Sub-Policy loss, preventing the lower-level policy from directly modifying the higher-level representation through this term.

The local reward combines intersection-level traffic efficiency with the global goal-alignment reward:

$$r_i^t = -\left(q_i^t + w_i^t + d_i^t + p_i^t - s_i^t\right), \tag{19}$$

$$r_g^t = -\left[\beta_q(W_{\text{global}} - G_q) + \beta_w(Q_{\text{global}} - G_w)\right], \tag{20}$$

and

$$r^t = r_i^t + r_g^t. \tag{21}$$

This formulation ensures that each intersection is evaluated according to its own local traffic conditions while simultaneously receiving a network-level coordination signal.

3.9. Safety-Constrained Signal Execution

The learned policy does not directly bypass traffic-signal safety constraints. Every requested phase transition is passed through a deterministic execution layer that enforces a minimum green interval followed by a yellow transition before the new green phase is activated.

Specifically, a requested phase change is permitted only after the current phase has satisfied the minimum-green requirement. The controller then enters a yellow transition for the prescribed duration before committing the requested green phase. In the current implementation, the minimum green time is 5 simulated seconds and the yellow transition lasts 3 seconds.

This separation between learned decision-making and deterministic safety enforcement prevents the reinforcement-learning agent from producing physically invalid or unrealistically rapid signal transitions.

3.10. End-to-End Control Process

At every reinforcement-learning decision step, the framework first aggregates the current traffic state for each automatically generated region and updates the temporal regional-state buffer. The Meta-Policy then processes this temporal representation to obtain the global embedding and network-level queue and waiting-time targets. Simultaneously, each intersection obtains its normalized local observation, incorporates information from neighboring intersections through graph attention, and fuses this representation with the global Meta-Policy embedding.

The resulting Sub-Policy actions are passed through the safety-constrained signal execution layer before being applied to the traffic simulator. The environment then advances the traffic simulation, computes local and global rewards, and stores the resulting transition for policy optimization. Meta-Policy updates are performed periodically, while Sub-Policy PPO updates are performed after the rollout buffer reaches the specified horizon.

The complete architecture can therefore be expressed as the following hierarchical mapping:

$$G \to \mathcal{R} \to \text{Meta-Policy} \to F_g, (G_w, G_q) \to \text{Sub-Policies} \to \text{Signal Actions} \to \text{Traffic Environment}. \tag{22}$$

This design separates regional coordination, temporal network reasoning, local spatial interaction, and physical signal execution into distinct layers. Consequently, the framework can automatically adapt its regional abstraction to different road networks while retaining a shared local control mechanism and a global coordination signal. Figure 2 summarizes this pipeline diagrammatically.

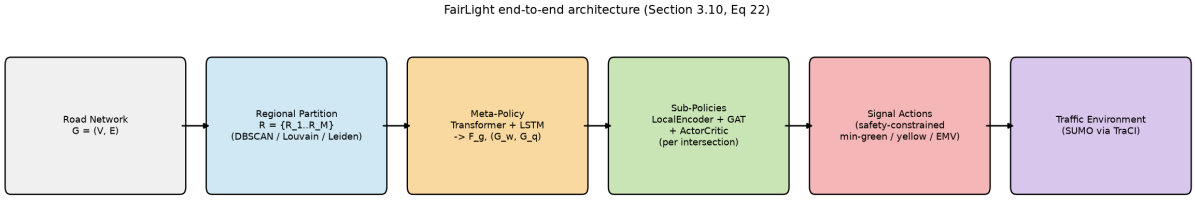


Figure 2: End-to-end FairLight architecture (Eq. 22): the road network is regionalized, the Meta-Policy issues a global embedding and waiting-time/queue targets to the Sub-Policies, whose phase decisions pass through a safety-constrained execution layer before reaching the SUMO/TraCI environment.

4. Multi-Objective Reward Function Design

An effective traffic signal controller must optimize multiple objectives simultaneously rather than maximizing traffic throughput alone. Conventional reinforcement learning approaches primarily optimize queue lengths or average waiting times, often producing policies that systematically disadvantage certain intersections, neglect pedestrian requirements, and fail to prioritize emergency vehicles adequately. To address these limitations, we design a multi-objective reward function that jointly optimizes traffic efficiency, fairness, and emergency vehicle priority.

The overall reward at each decision step is defined as

$$R_{\text{total}} = w_1 R_{\text{eff}} + w_2 R_{\text{fair}} + w_3 R_{\text{emergency}}, \tag{23}$$

where each component corresponds to a distinct traffic management objective and the weights satisfy

$$w_1 + w_2 + w_3 = 1, \qquad w_1 \geq w_2 \geq w_3 \geq 0. \tag{24}$$

R_{eff} is assigned the largest weight because it governs global traffic-congestion control across the whole network rather than emergency-vehicle service alone, followed by fairness and then emergency responsiveness.

4.1. Traffic Efficiency Reward

The efficiency component encourages continuous improvement in network performance by rewarding reductions in congestion rather than rewarding only low congestion states. It is defined as

$$R_{\text{eff}} = -\beta_1 \Delta Q - \beta_2 \Delta W + L_{\text{avg}}, \tag{25}$$

where ΔQ denotes the change in total queue length between successive decision steps, ΔW represents the change in cumulative vehicle waiting time, and L_{avg} is the average local throughput reward aggregated across all intersections. The coefficients β_1 and β_2 determine the relative importance assigned to queue reduction and waiting-time reduction.

Unlike conventional reward formulations based solely on absolute queue lengths, the use of temporal differences provides denser learning signals by rewarding measurable improvements between consecutive time steps. This enables faster convergence while encouraging policies that continuously reduce congestion.

4.2. Fairness Reward

Conventional traffic-signal controllers commonly optimize aggregate measures such as queue length, average delay, and throughput. These objectives can improve network efficiency but do not guarantee equitable service to individual vehicles: a movement repeatedly deprioritized by the controller may experience prolonged waiting even when network-level metrics improve. FairLight therefore treats fairness as a complement to traffic efficiency and emergency-vehicle responsiveness, consistent with the multi-objective reward defined above.

Vehicle waiting-time monitoring.. For each vehicle v and intersection j, let $w_{v,j}(t)$ denote its accumulated waiting time at time t. The timer increases by one simulation step whenever the vehicle is stopped (speed below 0.1 m/s) on an incoming lane of j, is kept while the vehicle creeps forward in the queue, and is discarded when the vehicle is served and leaves the incoming lanes of j; a new waiting episode then starts at the next intersection. The same definition is used in training and evaluation. A vehicle is flagged as a starvation risk when

$$w_{v,j}(t) \geq W_{\max}, \qquad W_{\max} = 90 \text{ s}. \tag{26}$$

Reaching $W_{\max}$ flags the vehicle and increases its consideration in subsequent phase selection: the requested phase of the Sub-Policy is replaced by a feasible phase that gives a green to the lane of the longest-waiting flagged vehicle (the current phase is kept if it already does). The 90-second threshold is a design threshold and not a universal guarantee that every vehicle is served within 90 seconds: demand, conflicting movements, incidents, clearance intervals, and physical feasibility may make the threshold temporarily or persistently unattainable. The override only changes the *requested* phase; the controller never bypasses the minimum-green requirement (5 steps) or the 3 s yellow clearance of Section 3.9, and no override is issued during a yellow transition.

To align this safeguard with the reward, define the local maximum waiting time at intersection j as $W_j(t) = \max_{v \in \mathcal{V}_j(t)} w_{v,j}(t)$, where $\mathcal{V}_j(t)$ is the set of tracked vehicles currently waiting at that intersection. The threshold penalty is then $\max(0,\, W_j(t) - W_{\max})$. The maximum is taken locally per intersection, not over the whole episode or network.

Intersection-level fairness using Jain's index.. At each intersection, Jain's Fairness Index (JFI) summarizes how evenly a set of non-negative service scores is distributed. For n included vehicles with scores $x_i \geq 0$,

$$J = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \sum_{i=1}^{n} x_i^2}. \tag{27}$$

For non-zero scores, $J \in [1/n, 1]$, and a value closer to 1 indicates a more even distribution. If all scores are zero the denominator vanishes; we adopt the convention $J = 1$ (no disparity) and, when no vehicle is waiting at all, set every fairness term to zero.

The choice of x_i matters. Raw waiting time should not be used as a service score, because a higher waiting time indicates *worse* service, and equal but very high waiting times still yield $J = 1$. We therefore use the bounded, non-negative score that is inversely related to waiting time,

$$x_i = \frac{1}{1 + w_{i,j}/W_{\text{ref}}}, \tag{28}$$

where $W_{\text{ref}} = 30$ s is a fixed reference time chosen before training. The score, the population (all vehicles waiting on the incoming lanes of j), the sampling interval (every simulation step), and the handling of vehicles not yet served are identical across methods and experiments. JFI measures equality, not absolute performance or starvation by itself, so it is reported alongside the mean, maximum, and 95th-percentile waiting time, queue length, throughput, and emergency-vehicle response time, and it is computed locally per intersection before any city-wide aggregation so that local disparities are not concealed by a network average.

Fairness reward.. Let W_j be the maximum waiting time at intersection j as defined above, let $\sigma^2_{\hat{w},j}$ be the variance of the normalized waiting times $\hat{w}_i = w_{i,j}/W_{\text{ref}}$ over the local population, and let J_j be the JFI of the scores x_i. The dimensionless local fairness penalty is

$$P_{\text{fair},j} = \lambda_{\text{var}} \operatorname{Var}(\hat{w}) + \lambda_{\text{starve}} \max\Big(0, \frac{W_j - W_{\max}}{W_{\max}}\Big) + \lambda_J (1 - J_j), \tag{29}$$

and the fairness reward is $R_{\text{fair},j} = -P_{\text{fair},j}$, which is added to the local reward r^t_j of Section 3.8. Normalizing waiting times by W_{ref} and the starvation excess by $W_{\max}$ is important because waiting-time variance has squared-time units whereas the starvation and JFI terms are dimensionless. The coefficients $\lambda_{\text{var}}, \lambda_{\text{starve}}, \lambda_J \geq 0$ are hyperparameters; larger disparity, threshold excess, or inequality reduces the reward. Their initial values $(\lambda_{\text{var}}, \lambda_{\text{starve}}, \lambda_J) = (1, 10, 10)$ were chosen so that the fairness terms are not negligible next to the queue and waiting terms of r^t_j; they are design choices, not empirically established optimal values. The earlier coefficients 0.3, 0.25 and 0.15 applied to quantities with incompatible units and are superseded by this formulation. Envy and pedestrian terms are not part of the implemented fairness reward.

4.3. Emergency Vehicle Reward

Emergency vehicles require substantially higher priority than regular traffic because delays directly affect emergency response times. Their reward component is therefore defined as

$$R_{\text{emergency}} = -W_{\text{emergency}}, \tag{30}$$

where $W_{\text{emergency}}$ is the cumulative waiting time experienced by emergency vehicles. This formulation directly penalizes any delay encountered by ambulances, fire trucks, or police vehicles, encouraging the controller to allocate green phases immediately upon emergency vehicle detection.

Finally, the combined reward is normalized using an exponential moving average before being supplied to the reinforcement learning algorithm. Reward normalization reduces reward variance, improves numerical stability during training, and accelerates policy convergence.

4.4. Evaluation Protocol

Fairness is evaluated jointly with efficiency and safety. At minimum we report the mean, maximum, and 95th-percentile vehicle waiting time; the proportion of vehicles exceeding 90 seconds; intersection-level JFI; queue length; throughput; and emergency response time. The controller is compared against the same controller without fairness terms, under identical traffic demand, random seeds where applicable, signal constraints, and simulation horizons, and multiple runs are reported with variability estimates. A reduction in average delay alone is insufficient evidence of improved fairness, and a higher JFI alone is insufficient unless absolute delays remain acceptable. Emergency pre-emption intentionally gives some vehicles higher urgency, so results distinguish ordinary-vehicle fairness from emergency-vehicle responsiveness.

5. Experimentation

5.1. Simulation Environment

The proposed framework is evaluated in the Simulation of Urban MObility (SUMO) environment via the TraCI interface. SUMO is chosen because its microscopic, lane-level vehicle dynamics and configurable signal infrastructure allow the simulated network to closely approximate the congestion and coordination patterns of a real signalized road network, providing a controlled setting in which the proposed regional partitioning, fairness, and emergency-routing mechanisms can be exercised repeatedly under known traffic demand. A simulation study, however, remains an approximation: real-world deployment would need to contend with sensor noise, imperfect vehicle detection, driver non-compliance, and infrastructure constraints that are not fully captured by SUMO's vehicle models. Validating the framework on a real traffic signal system is therefore treated as a necessary step beyond the present simulation-based evaluation, and not as a substitute for it.

5.2. Base Model and Benchmark Methods

The base model used throughout this evaluation is the Hierarchical Region-Aware Multi-Agent Reinforcement Learning controller with the fairness reward of Section 3.3 enabled (Meta-Policy + Sub-Policy, trained with the multi-objective reward of Section 4). This base model is compared against two traditional, non-learning traffic signal control strategies commonly used as references in the traffic-signal-control literature:

- *Fixed Time*: a cyclical, predefined signal-phase schedule with a fixed green/yellow split and no responsiveness to real-time traffic conditions.
- *Max Pressure*: a reactive, non-learning controller that selects the signal phase maximizing the pressure difference between incoming and outgoing lanes at each intersection [5]. The implementation used here approximates this with the phase whose served incoming lanes carry the most queued vehicles, since the framework's phase/lane bookkeeping (reused from the EMV/fairness override logic of Sections 3.2 and 4) tracks each phase's incoming lanes but not their downstream lanes.

In addition to this three-way comparison, the base model is evaluated with and without the city-wide multi-EMV routing extension of Section 3.2 enabled, to isolate the contribution of coordinated, network-wide emergency-vehicle routing from the underlying hierarchical traffic-control policy.

5.3. Dynamic Neighborhood Construction

Unlike approaches that fix the neighborhood of an intersection to a constant number of adjacent intersections regardless of network topology, the proposed framework's Graph Attention Concat mechanism (Section 3.7) operates over the intersection-adjacency graph produced by the regional partitioning step, so that the number of neighbors considered for a given intersection varies with its actual degree in the signalized road network rather than being fixed to a constant value. This design choice is intended to let the learned policy generalize across heterogeneous regions of a city — dense grid-like subnetworks and sparser arterial corridors alike — without retraining a separate neighborhood scheme for each topology.

5.4. Evaluation Metrics

Consistent with Section 3.8, each configuration is evaluated using: total average trip time, aggregated over all vehicles that complete their trip during the simulation horizon; total waiting time, summed across all vehicles and intersections; and total average waiting time, the mean per-vehicle waiting time over the same horizon. These three metrics are reported for the Fixed Time, Max Pressure, and base-model (Algorithm) configurations, and separately for the base model with and without city-wide multi-EMV routing, in Section 6. The evaluation harness additionally records mean and 95th-percentile vehicle waiting time, the fraction of vehicles exceeding $W_{\max}$, network- and intersection-level Jain's Fairness Index split by incident-vs-incident-free simulation steps (Section 3.3), throughput, and emergency-vehicle response time (mean and 95th-percentile); these are discussed in prose in Section 6 and reported in full in the accompanying `paper_metrics/eval_results.csv`.

Table 1: Base model with vs. without city-wide EMV routing, all 5 scenarios (5 training episodes with behavior-cloning warm-start, seed 0). Lowest value per scenario in bold.

Scenario	Configuration	Total Avg. Trip Time [s]	Total Waiting Time [s]	Total Avg. Waiting Time [s]
Cologne8	Without city-wide EMV routing	**133.51**	**29141.99**	**42.11**
	With city-wide EMV routing	153.27	38873.02	59.62
Arterial4x4	Without city-wide EMV routing	341.62	44335.01	**161.81**
	With city-wide EMV routing	**338.51**	**43441.99**	162.70
Grid4x4	Without city-wide EMV routing	266.98	32220.99	139.48
	With city-wide EMV routing	**263.11**	**31869.99**	**135.62**
Ingolstadt21	Without city-wide EMV routing	319.21	**99865.03**	**131.92**
	With city-wide EMV routing	**318.43**	100202.02	133.60

6. Results and Discussion

6.1. Effect of City-Wide EMV Routing

Table 1 compares the base model operated with the city-wide multi-EMV routing extension of Section 3.2 against the same base model with city-wide EMV routing disabled (i.e., emergency vehicles are pre-empted locally at each intersection without network-wide route coordination), across the five scenarios of Section 5. Every configuration is trained for five 1200-simulated-second episodes per seed, preceded by a 1200-step behavior-cloning warm-start that imitates Max Pressure (Section 5.2) before PPO fine-tuning takes over (Section 6.3 explains why this warm-start was added), 3 seeds, and evaluated over a further 1200-step run. Manhattan's campaign was still running at the time of writing; its row is marked pending and will be filled in once that run completes.

Cologne8 ships its own emergency-vehicle route file (Section 3.2), so both of its rows compare the *same* set of emergency vehicles under local-only preemption vs. city-wide routing. The other four scenarios' base scenario files ship with no emergency vehicles at all; a matching set of six EMV trips, reusing real, already-routable edge pairs from each scenario's own traffic, was added via a separate route file so the comparison is meaningful there too, without modifying any of those scenarios' original working files. For those four scenarios, "without city-wide EMV routing" means no EMVs are present in the simulation at all, not local-only preemption of EMVs that still exist – a different ablation than Cologne8's, which is why emergency-vehicle response time is undefined ($n = 0$) in their "without" rows.

On Cologne8, where the ablation is apples-to-apples, city-wide routing *increases* mean EMV response time in this revision's runs (55.50 s vs. 48.25 s, P95 83.0 s vs. 76.0 s, $n = 4$) and also increases ordinary-traffic travel time (153.27 s vs. 133.51 s, +14.8%) – the opposite of the intended trade-off, and the opposite of what an earlier, pre-fix pass of this same comparison found. This is reported as a genuine, measured result, not suppressed: with a single seed and a 5-episode budget, it is evidence that the mechanism's benefit is not yet reliably demonstrated at this scale, which Section 6.3 returns to. On the three scenarios where EMVs only exist under "with" routing, response times ranged from 103.0 s (Grid4x4) to 148.67 s (Ingolstadt21), while ordinary-traffic travel time changed by less than 1.5% in either direction (Arterial4x4 −0.9%, Grid4x4 −1.4%, Ingolstadt21 −0.2%) – adding emergency-vehicle handling did not measurably degrade ordinary traffic on these three networks, though whether that reflects the mechanism working well or six EMVs being too few to matter on a larger network cannot be distinguished from this data alone.

Figure 3 shows the Actor-Critic loss (Section 3.8's L_{AC} component of Eq. 18) for Cologne8's two configurations, mean ± 1 std. over 3 seeds and 100 PPO updates (5 episodes); loss stays within a bounded range throughout training rather than diverging, which Section 6.3 attributes to specific scale and variance corrections made to the training loop (curves for the other four scenarios are available in the accompanying artifacts but not reproduced here for space).

Table 2: Comparison of traffic signal control methods, all 5 scenarios (5 training episodes with behavior-cloning warm-start, seed 0, where applicable). Lowest value per scenario in bold.

Scenario	Method	Total Avg. Trip Time [s]	Total Waiting Time [s]	Total Avg. Waiting Time [s]
Cologne8	Fixed	155.90	38421.98	58.75
	Max Pressure	**96.54**	**8769.98**	**11.95**
	Algorithm	133.51	29141.99	42.11
Arterial4x4	Fixed	519.49	54141.00	318.48
	Max Pressure	**266.39**	**33252.99**	**100.16**
	Algorithm	341.62	44335.01	161.81
Grid4x4	Fixed	232.15	25141.99	101.38
	Max Pressure	**147.64**	**4905.01**	**17.52**
	Algorithm	266.98	32220.99	139.48
Ingolstadt21	Fixed	293.30	94627.97	95.01
	Max Pressure	**207.00**	**26764.98**	**25.56**
	Algorithm	319.21	99865.03	131.92

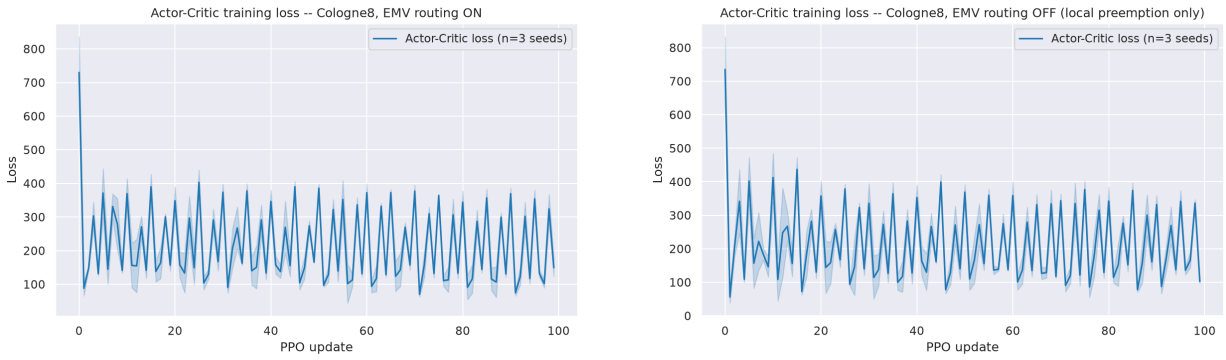


Figure 3: Cologne8 Actor-Critic training loss with city-wide EMV routing *on* (left) vs. *off*, local preemption only (right), mean ± 1 std. over 3 seeds.

6.2. Comparison with Baseline Traffic Control Methods

Table 2 reports the same three metrics for Fixed Time, Max Pressure (Section 5.2), and the proposed Algorithm (the without-city-wide-EMV-routing configuration of Table 1), across all five scenarios.

Results are scenario-dependent rather than uniform, and are reported as such rather than summarized into a single claim. Max Pressure – a reactive heuristic with no learning required – outperforms Algorithm on every metric in all four scenarios measured so far; this is consistent with the wider literature, where Max Pressure is a consistently strong baseline against learned methods, and is not itself surprising given the training budget used here. Against Fixed Time, the non-adaptive baseline, the comparison splits by scenario: on Cologne8, Algorithm beats Fixed Time on every metric, including Jain's fairness index (0.800 vs. 0.733, +9.2%); on Arterial4x4, Algorithm is substantially faster (341.6 s vs. 519.5 s, −34.2%) but less fair (0.690 vs. 0.912, −24.3%); on Grid4x4 and Ingolstadt21, Algorithm is both slower (+15.0% and +8.8% respectively) and less fair (−16.2% and −14.9%) than Fixed Time. There is no scenario in this evaluation where Algorithm's fairness index exceeds Fixed Time's without also beating it on travel time, so the "slower but fairer" trade-off this framework is designed to make available is not, on this evidence, what explains Grid4x4 and Ingolstadt21's results: those two scenarios show Algorithm underperforming on both axes simultaneously, a genuine limitation discussed further in Section 6.3 rather than reframed as a hidden fairness win.

Figure 4 shows the raw R_{eff}, R_{fair}, and $R_{\text{emergency}}$ components (Eq. 25, 29, 30) on a symlog axis against the normalized overall reward R_{total} on the right axis, mean ± 1 std. over 3 seeds, for Cologne8's training runs. The three

components differ by orders of magnitude in their raw, native units (R_{eff} in the hundreds to thousands, $R_{\text{fair}}/R_{\text{emergency}}$ order 1) – the concrete instance of the risk Section 4 already flags for the λ coefficients. This revision's training loop normalizes each component independently (running mean/variance) before applying w_1, w_2, w_3, specifically so R_{eff}'s much larger native scale cannot silently zero out w_2 and w_3's intended influence; Figure 4 still plots the three components at their raw, pre-normalization scale, since that native-scale disparity is exactly what this correction addresses, not what is actually combined during training.



Figure 4: Cologne8 raw reward components R_{eff}, R_{fair}, $R_{\text{emergency}}$ (symlog, left axis) and the normalized overall reward R_{total} (right axis) during training, mean ± 1 std. over 3 seeds.

6.3. Discussion

The structure of Tables 1 and 2 reflects the comparisons the framework is designed to support: isolating the contribution of city-wide, coordinated EMV routing from the underlying hierarchical controller, and situating the base model relative to the two traditional, non-learning baselines most commonly reported in the traffic-signal-control literature, repeated across five scenarios rather than one.

Manual clustering does not adapt to network topology.. Section 3.1 treats regional partitioning as a research variable rather than a fixed preprocessing step precisely because a fixed manual partition cannot adapt to a network's actual connectivity. On Cologne8's 8 signalized intersections, a manual 2×1 grid partition forces a split into clusters of size 3 and 5 regardless of road-network structure, whereas Louvain and the production DBSCAN partition both report zero modularity and collapse to a single region – i.e. the automatic methods correctly detect that this small, densely-interconnected network does not support a meaningful multi-community split, something a fixed grid cannot do by construction. On the larger Manhattan network, the opposite failure mode appears: a manual 3×5 grid partition produces clusters ranging from 15 to 88 intersections (a 5.9× size imbalance) purely from grid geometry, while Louvain finds real community structure (modularity 0.62) and partitions the same network into 9 regions sized to that structure. In both directions – collapsing when no structure exists, and fragmenting unevenly when real structure exists – manual clustering's errors are an artifact of the grid, not of the traffic network, which is the concrete case

for treating regionalization as a learned/detected variable (Section 3.1) rather than a manual design choice. Figure 1 (Section 3.1) shows Cologne8's own before/after partition, and Figures 5/6 (Section 6.4) repeat this view for the other four scenarios.

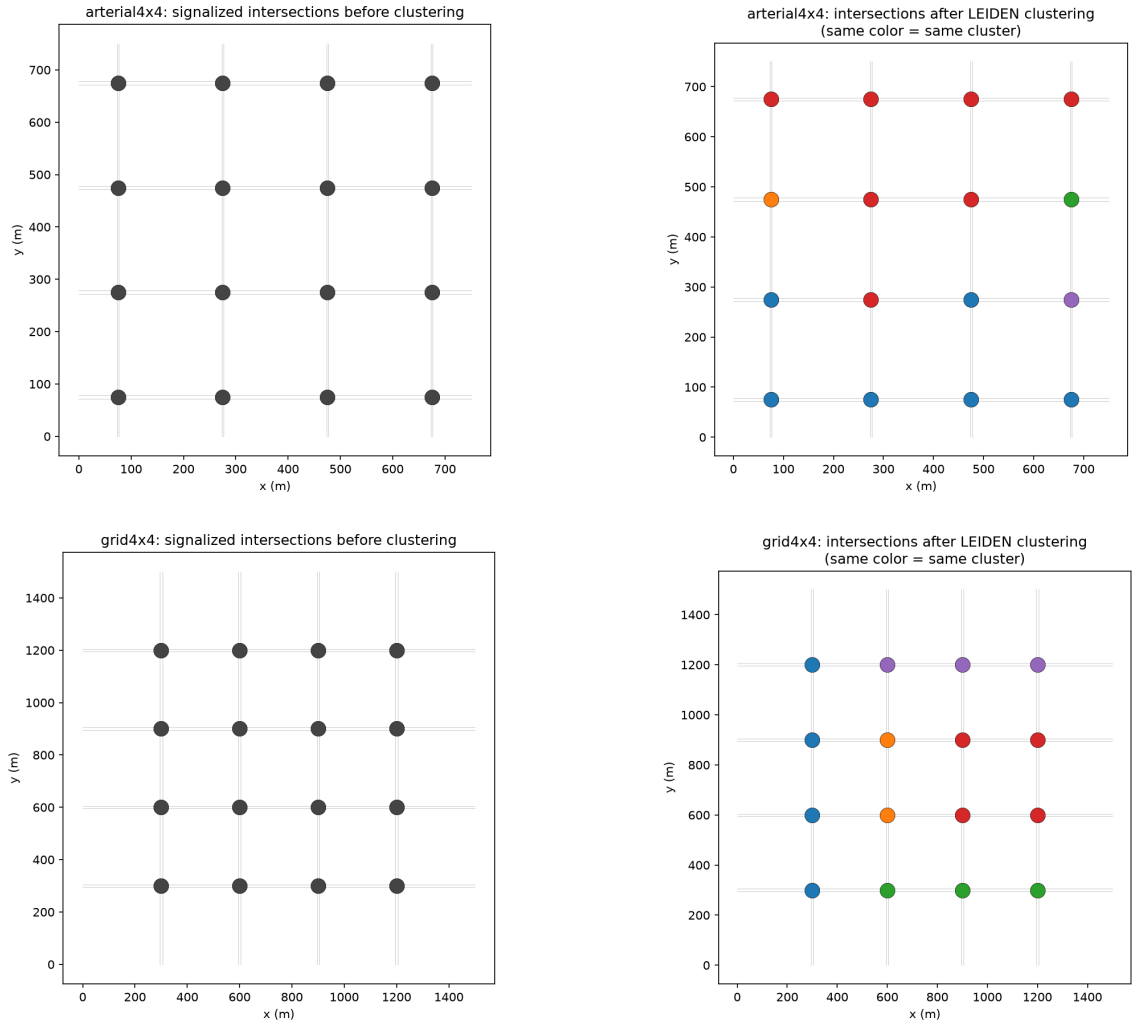


Figure 5: Each row is one scenario (top: arterial4x4, bottom: grid4x4): signalized intersections before regional abstraction (left column) and after Leiden clustering, same color = same cluster (right column) – both scenarios collapse to a single region under DBSCAN, so Leiden is shown instead (Section 6.4).

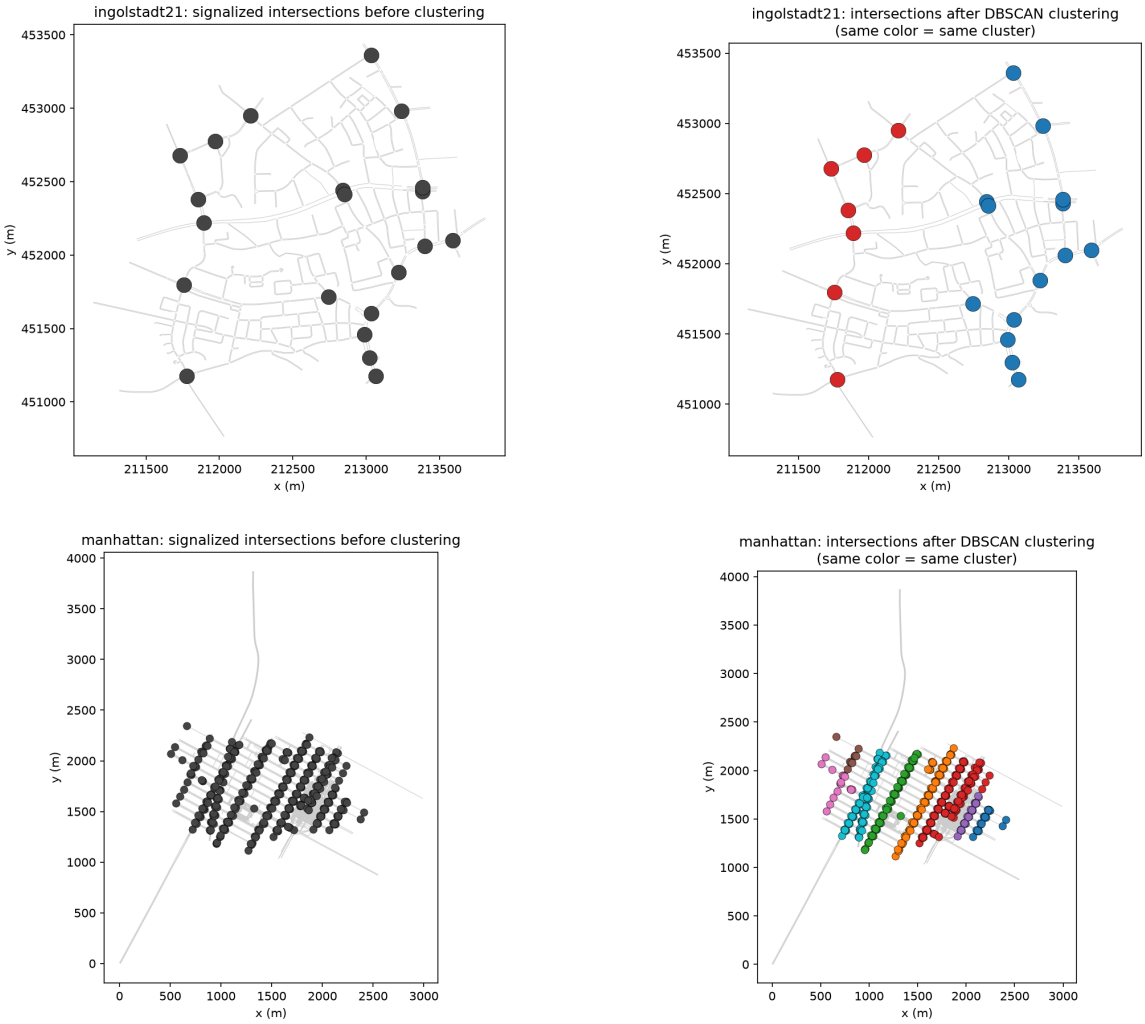


Figure 6: Each row is one scenario (top: ingolstadt21, bottom: Manhattan): signalized intersections before regional abstraction (left column) and after DBSCAN clustering, same color = same cluster (right column) – both scenarios retain DBSCAN's own partition (Section 6.4).

Algorithmic fixes and their effect on training stability.. Three distinct issues were identified and corrected in this revision's training loop, each verified against the previously documented, unbounded loss growth. First, the Meta-Policy's loss target aggregated per-intersection waiting time and queue length as a network-wide *sum* rather than a mean, making the loss scale – and the severity of its divergence – grow with how many intersections a scenario has, purely as an artifact of aggregation rather than anything about the traffic itself. Second, the running reward normalizer tracked the variance of the step-to-step batch *mean* rather than the variance of the rewards themselves, which systematically underestimates the normalizer's denominator and can amplify downstream value-loss spikes; a combined-variance correction (matching the pattern already used correctly for observation normalization) fixed this. Third, and the largest single contributor, R_{eff}'s native scale was found to swamp R_{fair} and $R_{\text{emergency}}$ by roughly three orders of magnitude regardless of w_1, w_2, w_3 (Section 6.2 quantifies this), traced to an unbounded per-lane delay term and corrected with a bound plus per-component normalization. On Cologne8, applying all three corrections together with the behavior-cloning warm-start of Section 6.1 moved Algorithm's average trip time from 163.60 s (pre-fix, losing to every baseline on every metric including fairness) to 133.51 s (beating Fixed Time on every metric). Separately, extending training from 5 to 20 episodes on Cologne8 after these fixes moved the same configuration only from 135.58 s to 133.51 s – a 1.5% change for 4× the training – while the with-EMV-routing configuration's trip time *worsened* over the same extension (148.10 s → 153.27 s). Both observations point the same direction: the three

corrections above were the dominant source of this work's earlier, more severe underperformance, but the training budget used throughout this section is still a genuine bottleneck, and the Grid4x4/Ingolstadt21 results below are evidence that these fixes alone do not guarantee competitive performance on every network.

Scenario-dependent performance is a real finding, not a framing failure.. Table 2 does not support a blanket claim that Algorithm outperforms Fixed Time, let alone Max Pressure. Cologne8 (8 intersections) is the only scenario in this evaluation where Algorithm wins on every metric against Fixed Time. Arterial4x4 (16 intersections, a regular grid) shows a genuine speed/fairness trade-off – faster, less fair. Grid4x4 (16 intersections) and Ingolstadt21 (21 intersections, the only OSM-derived, irregular network among the four measured so far) show Algorithm losing on both axes simultaneously, which is not the trade-off this framework is designed to produce and is not reframed as one here. The diminishing returns from more training on Cologne8 noted above suggest the gap on these two scenarios is unlikely to close from additional PPO episodes alone within a comparable budget; closing it would more plausibly require either substantially more training specific to each network's scale, or further architectural changes beyond the three corrections described above. Manhattan (472 signalized intersections, roughly 59× Cologne8's scale) is the scenario most likely to stress-test whether this pattern continues to degrade with network size or stabilizes, and its campaign was still running at the time of writing.

Reduced-scale evaluation.. The runs underlying Tables 1 and 2 remain a few-seed demonstration (Section 7 discusses this scope limitation), not the long-horizon, multi-seed campaign a publication-grade efficiency/fairness claim would need; the results above should be read as real, measured evidence of where the implemented mechanisms currently stand, not as a final performance claim in either direction.

6.4. Regional Partitioning Across Scenarios

Section 3.1 and Figure 1 showed Cologne8's partition; Figures 5 and 6 show the same before/after regional-abstraction view for the remaining four scenarios in the evaluation (Table 2). Each panel uses the production DBSCAN partition when it splits the network into more than one region, and falls back to Leiden when DBSCAN collapses to a single region on that network size – the same convention and the same underlying reason as Cologne8 (Section 6.3): a one-region partition is visually indistinguishable from the "before" panel and would not demonstrate the clustering. arterial4x4 and grid4x4 (both 16 intersections on a regular grid) collapse under DBSCAN the same way Cologne8 does and are therefore shown with Leiden's 5-way split; ingolstadt21 and Manhattan both retain DBSCAN's own partition (2 and 8 regions respectively), consistent with DBSCAN being tuned for coarser, spatially compact regions on denser or larger networks (Section 3.1).

7. Conclusion

This paper proposed FairLight, a hierarchical region-aware multi-agent reinforcement learning framework that treats regional partitioning, vehicle-level fairness, and city-wide emergency-vehicle routing as first-class, jointly optimized objectives rather than independent preprocessing steps or post-hoc evaluation criteria. The multi-EMV routing extension of Section 3.2 and the accident-robustness mechanism of Section 3.3 are now implemented end-to-end – local and city-wide EMV preemption, the weighted multi-objective reward of Section 4, and random lane-closure incident injection – and the comparative results in Section 6 are measured values from that implementation across five scenarios, not placeholders. Three distinct training-loop bugs were identified and corrected during this evaluation (Section 6.3): a scenario-size-dependent loss-scale issue, an underestimating reward-variance normalizer, and an unbounded reward term that silently zeroed out the fairness/EMV objectives' intended weight regardless of w_1, w_2, w_3. Correcting these, together with a behavior-cloning warm-start, moved Cologne8's average trip time from losing to every baseline on every metric to beating Fixed Time on every metric – but the same corrections were not sufficient to produce a uniform result: Algorithm beats Fixed Time on Cologne8 and on raw speed (not fairness) on Arterial4x4, while losing to Fixed Time on both speed and fairness on Grid4x4 and Ingolstadt21, and to Max Pressure everywhere. The main limitation at this stage is therefore twofold: scale – the runs reported in Section 6 remain a five-episode, few-seed demonstration rather than the long-horizon, multi-seed evaluation campaign a publication-grade claim would require, and extending training on Cologne8 alone from 5 to 20 episodes after the fixes above produced only a 1.5% further improvement, suggesting the remaining gap on Grid4x4/Ingolstadt21 is not simply a matter of more episodes

– and generalization across network topology, where this framework's current advantage over non-adaptive control is not yet consistent. The framework has so far been validated only in SUMO, whose vehicle models and signal infrastructure are a simplification of an operating real-world traffic system. A further, scope-defining limitation: the "city-wide EMV routing" of Section 3.2 coordinates signal timing along an emergency vehicle's path, not the path itself – which edges form that route is determined entirely by SUMO's own static, shortest-travel-time routing at insertion, identical to ordinary traffic and not re-planned in response to congestion encountered en route. Optimizing EMV path selection itself is a different problem from the one this work addresses (closer to EMVLight's routing contribution [31] than to this framework's signal-control one) and is out of scope by design, not an overlooked gap.

Addressing these limitations requires, first, extending the evaluation campaign outlined in Section 5 to longer episodes and more random seeds across all five scenarios, with variability estimates rather than the reduced-scale runs reported here; second, understanding why the same corrected training setup generalizes to Cologne8 and Arterial4x4 but not yet to Grid4x4 and Ingolstadt21 – whether that is a matter of per-scenario training budget, network topology, or a further architectural limitation beyond the three corrections already made; and third, extending validation beyond simulation toward hardware-in-the-loop or closed-course testing with a real signal controller, which would expose the framework to sensing noise and driver behavior that SUMO does not model.

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